Soutenance

NAS Debian

Veille

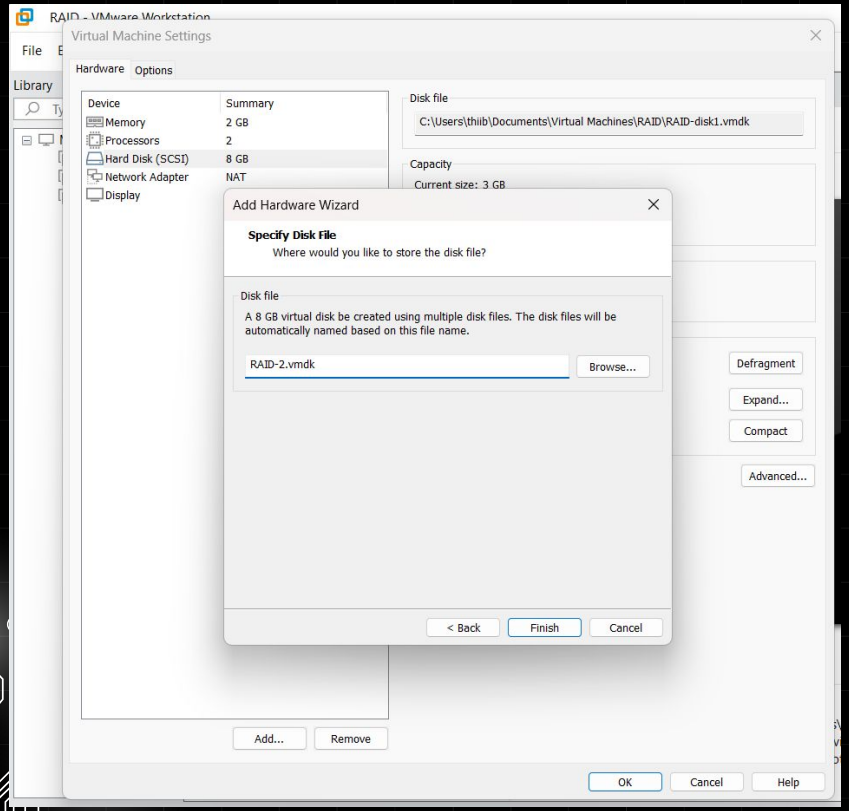
Un serveur NAS : (Network Attached Storage) est un appareils de stockage autonome qui peut se connecter à votre réseau privé ou professionnel via internet. Il permet de sauvegarder,partager et sécuriser mais aissi faciliter l'accès à vos fichiers depuis plusieurs appareils.

Un RAID : (Redundant Array of Independent Disks) est un type stockage consistant à écrire des données sur plusieurs disques d’un même système.

WebDAV : est un protocole qui permet de gérer les fichiers avec des serveurs distants.

SFTP : est aussi un protocole réseau standard pour le transfert sécurisé de fichiers entre les systèmes connectés.

Construction du RAID



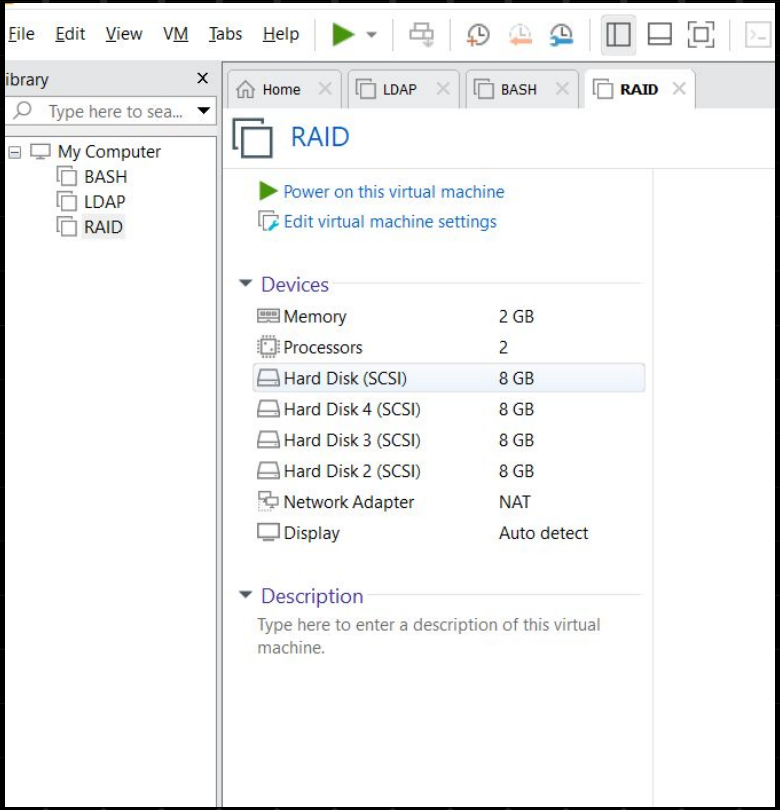

```
root@debian-62f49bb4:~# mdadm --create /dev/md0 --level=5 --raid-devices=3 /
dev/sdb /dev/sdc /dev/sdd
To optimalize recovery speed, it is recommended to enable write-indent bitma
p, do you want to enable it now? [y/N]? y
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md0 started.
root@debian-62f49bb4:~# cat /proc/mdstat
Personalities : [raid0] [raid1] [raid6] [raid5] [raid4] [raid10]
md0 : active raid5 sdd[3] sdc[1] sdb[0]
      16758784 blocks super 1.2 level 5, 512k chunk, algorithm 2 [3/2] [UU_]
      [==>..................]  recovery = 13.2% (1107836/8379392) finish=0.7
min speed=158262K/sec
      bitmap: 1/1 pages [4KB], 65536KB chunk

unused devices: <none>
root@debian-62f49bb4:~# cat /proc/mdstat
Personalities : [raid0] [raid1] [raid6] [raid5] [raid4] [raid10]
md0 : active raid5 sdd[3] sdc[1] sdb[0]
      16758784 blocks super 1.2 level 5, 512k chunk, algorithm 2 [3/2] [UU_]
      [========>............]  recovery = 40.7% (3414972/8379392) finish=0.4
min speed=200880K/sec
      bitmap: 0/1 pages [0KB], 65536KB chunk

unused devices: <none>
root@debian-62f49bb4:~# cat /proc/mdstat
Personalities : [raid0] [raid1] [raid6] [raid5] [raid4] [raid10]
md0 : active raid5 sdd[3] sdc[1] sdb[0]
      16758784 blocks super 1.2 level 5, 512k chunk, algorithm 2 [3/2] [UU_]
      [==================>..]  recovery = 91.4% (7664220/8379392) finish=0.0
min speed=205597K/sec
      bitmap: 0/1 pages [0KB], 65536KB chunk

unused devices: <none>
root@debian-62f49bb4:~#
```

Construction du RAID



Construction du RAID (finalisation)

```
root@debian-62f49bb4:~# cat /proc/mdstat
Personalities : [raid0] [raid1] [raid6] [raid5] [raid4] [raid10]
md0 : active raid5 sdd[3] sdc[1] sdb[0]
      16758784 blocks super 1.2 level 5, 512k chunk, algorithm 2 [3/3] [UUU]
      bitmap: 0/1 pages [0KB], 65536KB chunk

unused devices: <none>
```

```
root@debian-62f49bb4:~# df -h | grep md0
/dev/md0            16G    2,1M   15G   1% /mnt/raid5
root@debian-62f49bb4:~#
```

Le SFTP

```
laplateforme@debian-62f49b

  GNU nano 8.4                    /etc/ssh/sshd_config
#X11UseLocalhost yes
#PermitTTY yes
PrintMotd no
#PrintLastLog yes
#TCPKeepAlive yes
#PermitUserEnvironment no
#Compression delayed
#ClientAliveInterval 0
#ClientAliveCountMax 3
#UseDNS no
#PidFile /run/sshd.pid
#MaxStartups 10:30:100
#PermitTunnel no
#ChrootDirectory none
#VersionAddendum none

# no default banner path
#Banner none

# Allow client to pass locale and color environment variables
AcceptEnv LANG LC_* COLORTERM NO_COLOR

# override default of no subsystems
Subsystem       sftp    /usr/lib/openssh/sftp-server

# Example of overriding settings on a per-user basis
#Match User anoncvs
#       X11Forwarding no
#       AllowTcpForwarding no
#       PermitTTY no
#       ForceCommand cvs server

Match User Bob
    ChrootDirectory /mnt/raid5
    ForceCommand internal-sftp
    AllowTcpForwarding no

^G Aide        ^O Écrire      ^F Chercher    ^K Couper      ^T Exécuter
^X Quitter     ^R Lire fich.  ^\ Remplacer   ^U Coller      ^J Justifier
```

```
C:\Users\thiib>sftp Bob@192.168.211.133
Bob@192.168.211.133's password:
Connected to 192.168.211.133.
sftp> ls
lost+found  sftp
sftp> pwd
Remote working directory: /
sftp>
```

le SFTP (avec les autres utilisateurs)

```
C:\Users\thiib>sftp michel@192.168.211.133
michel@192.168.211.133's password:
Connected to 192.168.211.133.
sftp> ls
lost+found  public      sftp        users       webdav
sftp> cd users/michel
sftp> pwd
Remote working directory: /users/michel
sftp> put C:\Users\thiib\test.txt
Uploading C:/Users/thiib/test.txt to /users/michel/test.txt
test.txt                                      100%   11     2.7KB/s   00:00
sftp> exit

C:\Users\thiib>sftp rihana@192.168.211.133
rihana@192.168.211.133's password:
Connected to 192.168.211.133.
sftp> pwd
Remote working directory: /
sftp> ls
lost+found  public      sftp        users       webdav
sftp> cd users/rihana
sftp> pwd
Remote working directory: /users/rihana
sftp> exit

C:\Users\thiib>sftp alicia@192.168.211.133
alicia@192.168.211.133's password:
Connected to 192.168.211.133.
sftp> ls
lost+found  public      sftp        users       webdav
sftp> cd users/alicia
sftp> pwd
Remote working directory: /users/alicia
sftp> exit

C:\Users\thiib>
```

SFTP (preuve de transfert)

```
sftp> put C:\Users\thiib\test.txt
Uploading C:/Users/thiib/test.txt to /sftp/test.txt
test.txt                                      100%   11     2.2KB/s   00:00
sftp>
```

```
root@debian-62f49bb4:~# ls /mnt/raid5/sftp/
test.txt
root@debian-62f49bb4:~#
```

WebDav

```
root@debian-62f49bb4:~# mkdir -p /mnt/raid5/webdav
root@debian-62f49bb4:~# chown -R www-data:www-data /mnt/raid5/webdav
root@debian-62f49bb4:~# nano /etc/apache2/sites-available/webdav.conf
root@debian-62f49bb4:~# a2ensite webdav.conf
Enabling site webdav.
To activate the new configuration, you need to run:
  systemctl reload apache2
root@debian-62f49bb4:~# systemctl restart apache2
root@debian-62f49bb4:~# a2ensite webdav.conf
Site webdav already enabled
root@debian-62f49bb4:~# a2dissite 000-default.conf
Site 000-default disabled.
To activate the new configuration, you need to run:
  systemctl reload apache2
root@debian-62f49bb4:~# systemctl restart apache2
root@debian-62f49bb4:~# a2dissite 000-default.conf
Site 000-default already disabled
root@debian-62f49bb4:~# systemctl restart apache2
root@debian-62f49bb4:~# echo "Je suis professionnelle" > /mnt/raid5/webdav/t
est.txt
root@debian-62f49bb4:~#
```

WebDav

```
  GNU nano 8.4      /etc/apache2/sites-available/webdav.conf
<VirtualHost *:80>
    ServerName 192.168.211.133
    DocumentRoot /mnt/raid5/webdav

    <Directory /mnt/raid5/webdav>
        DAV On
        Options Indexes
        AllowOverride None
        Require all granted
    </Directory>
</VirtualHost>
```

WebDav

Index de /

Nom	Dernière modification	Taille	Description
test.txt	2026-03-04 12:17	24	

Serveur Apache/2.4.66 (Debian) à l'adresse IP 192.168.211.133, port 80

Non sécurisé 192.168.211.133/test.txt

Scolaire

```
Je suis professionnelle
```

WebDav

```
  GNU nano 8.4      /etc/apache2/sites-available/webdav.conf
<VirtualHost *:80>
    ServerName 192.168.211.133
    DocumentRoot /mnt/raid5/webdav

    <Directory /mnt/raid5/webdav>
        DAV On
        Options Indexes
        AllowOverride None
        Require all granted
    </Directory>

    Alias /public /mnt/raid5/public
    <Directory /mnt/raid5/public>
        DAV On
        Options Indexes
        AllowOverride None
        Require all granted
    </Directory>
</VirtualHost>
```

WebDav

Index of /public

Name	Last modified	Size	Description
Parent Directory		-	
test2.txt	2026-03-04 12:29	30	

Apache/2.4.66 (Debian) Server at 192.168.211.133 Port 80

Non sécurisé 192.168.211.133/public/test2.txt

Scolaire

```
je suis trÃ©s professionnelle
```

Difficultés rencontrées

Le SFTP : Tentative de connexion à l'utilisateur (connexion closed).

WebDav : Manque de veille et manque de compréhension du but de WebDav.

Nous l'avons fait dans le mauvais sens, c'est à dire qu'on a tout créé avant de faire le RAID et donc, nous avons dû déplacer le /home.

Le sujet était vaste pas précis, manque de directive.

FIN